claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a2d2f44d7dd9742e4.jsonl`
- SHA-256: `81986e18ccf62fc1f02d955bfeb4f011dc015b1e31a4357e5186784aa165839b`
- Source modified: `2026-05-31T04:26:17+00:00`
- Imported at: `2026-07-05T16:48:20+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:25:15.653Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On table extraction accuracy (table edit-distance, lower is better), the PP-StructureV3 OCR+table-structure pipeline outperforms large general-purpose VLMs and pipeline competitors: it scores 0.159 (EN) / 0.109 (ZH) on Table Edit, versus Qwen2.5-VL-72B at 0.341/0.262, GOT-OCR at 0.459/0.52, Marker-1.2.3 at 0.619/0.685, MinerU-0.9.3 at 0.18/0.344, and Mathpix at 0.243/0.32. This supports a deterministic-pipeline-over-VLM recommendation for dense tables."

Source: https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md (primary)
Supporting quote: "PP-StructureV3 scores: ... Table Edit: EN 0.159, ZH 0.109 ... Qwen2.5-VL-72B: ... Table 0.341 (EN), 0.262 (ZH) ... GOT-OCR: ... Table 0.459 (EN), 0.52 (ZH) ... Marker-1.2.3: ... Table 0.619 (EN), 0.685 (ZH)"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:25:18.590Z)

I'll verify this claim about PP-StructureV3's table extraction benchmarks. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-05-31T04:25:21.776Z)

Let me fetch the primary source and search for independent verification in parallel.

4. user (2026-05-31T04:25:28.805Z)

PP-StructureV3 Table Edit Benchmark Results

Metric Details
The benchmark uses **"TableEdit?"** scores, where **lower values are better** (it measures edit distance/error rate). The data comes from **OmniDocBench**, specifically the paper: *"OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations"* available at https://arxiv.org/abs/2412.07626.

Table Edit Distance Scores

Method	EN	ZH
PP-StructureV3	**0.159**	**0.109**
Qwen2.5-VL-72B	0.341	0.262
GOT-OCR	0.459	0.52
Marker-1.2.3	0.619	0.685
MinerU-1.3.11	0.1613	0.1833
Mathpix	0.243	0.32

PP-StructureV3 achieves the lowest (best) table edit distance scores among all methods tested, indicating superior table recognition performance across both English and Chinese documents.

5. user (2026-05-31T04:25:33.113Z)

Web search results for query: "PP-StructureV3 OmniDocBench table edit distance Qwen2.5-VL-72B GOT-OCR MinerU benchmark"

Links: [{"title":"GitHub - marklabz/omnidocbench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub","url":"https://github.com/marklabz/omnidocbench"},{"title":"GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation","url":"https://github.com/opendatalab/OmniDocBench?tab=readme-ov-file"},{"title":"PaddleOCR/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md at main · PaddlePaddle/PaddleOCR","url":"https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md"},{"title":"GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub","url":"https://github.com/opendatalab/OmniDocBench"},{"title":"OmniDocBench v1.0 Benchmark","url":"https://www.emergentmind.com/topics/omnidocbench-v1-0"},{"title":"MinerU2.5: A Decoupled Vision-Language Model for Efficient","url":"https://wangbindl.github.io/publications/MinerU2_5.pdf"},{"title":"PP-StructureV3 Introduction - PaddleOCR Documentation","url":"https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"},{"title":"OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA","url":"https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"},{"title":"Real5-OmniDocBench: A Full-Scale Physical Reconstruction Benchmark for Robust Document Parsing in the Wild","url":"https://arxiv.org/html/2603.04205v1"}]

Based on the search results, here's what I found about these models and benchmarks:

OmniDocBench Overview

OmniDocBench includes 1355 PDF pages, covering 9 document types, 4 layout types, and 3 language types, and features rich annotations, containing 15 block-level annotations (text paragraphs, headings, tables, etc.) and 4 span-level annotations (text lines, inline formulas, subscripts, etc.)

Edit Distance and Table Evaluation

Edit Distance calculation for the Table section has been updated to use normalized GTs and

Preds. Tree Edit Distance Score is used for table structure.

Model Evaluations

Qwen2.5-VL-72B model evaluation was added in March 2025. According to benchmark results, Qwen2.5-VL (72B) achieved 87.02 for one metric with 0.094 table edit distance, while PP-StructureV3 achieved 86.73 with 0.073 table edit distance.

PP-StructureV3, MinerU2.0-vlm, and Qwen2.5-VL-7B model evaluations were added in July 2025. Additionally, MinerU2.5 outperforms both general-purpose VLMs (e.g., Gemini-2.5 Pro, Qwen2.5-VL-72B, GPT-4o) and domain-specific models (e.g., PP-StructureV3).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:25:35.979Z)

I notice an important discrepancy already. The claim cites MinerU-0.9.3 at 0.18/0.344, but the current primary source shows MinerU-1.3.11 at 0.1613/0.1833. The search results also surface conflicting numbers from different OmniDocBench versions and a competing claim from MinerU2.5. Let me dig deeper.

7. user (2026-05-31T04:25:39.296Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html
Redirect URL: http://www.paddleocr.ai/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html
Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "http://www.paddleocr.ai/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"
- prompt: "Find the OmniDocBench benchmark comparison table. List the exact "Table Edit" or "TableEdit" scores (EN and ZH) for every method shown, especially: PP-StructureV3, Qwen2.5-VL-72B, GOT-OCR, Marker (with version), MinerU (with version number), Mathpix. Quote verbatim. Does it mention "MinerU-0.9.3" anywhere or a different MinerU version? What does the page say about how PP-StructureV3 compares to VLMs?"

8. user (2026-05-31T04:25:46.216Z)

OmniDocBench Table Recognition Results

I cannot find a dedicated "Table Edit Distance" leaderboard in the provided content. However, the document includes a **Table Recognition Evaluation** section showing TEDS scores (not edit distance for tables).

Available Table Metrics

The table recognition evaluation shows **TEDS and Edit_dist metrics**, but the leaderboard provided only displays TEDS results. For the models you mentioned:

From the Table Recognition section:
- "RapidTable: 82.5" (overall TEDS)
- "GOT-OCR: 74.9" (overall TEDS)
- "Qwen2-VL-7B: 71.0" (overall TEDS)

Missing Information

PP-StructureV3, Qwen2.5-VL-72B, Mathpix, and MinerU variants do **not appear** in the table recognition results table shown. The End-to-End leaderboard includes these models with different metrics (overall scores combining text, formula, and table results), but not isolated table edit distance comparisons.

The document notes that table evaluation uses both TEDS and Edit_dist metrics in configuration, but the published results table only reports TEDS scores.

9. assistant (2026-05-31T04:25:50.706Z)

The MinerU version discrepancy is significant ? the claim's "MinerU-0.9.3 at 0.18/0.344" matches neither the current PaddleOCR doc (MinerU-1.3.11 at 0.1613/0.1833) nor the official OmniDocBench leaderboard. This strongly suggests the claim was drawn from an older snapshot of the PaddleOCR doc. Let me confirm the current doc state and check the substantive interpretation issue (whether PP-StructureV3 actually beats the strongest competitors).

10. user (2026-05-31T04:25:57.743Z)

OmniDocBench TableEdit Scores Analysis

Based on the documentation provided, here are the TableEdit scores from the benchmark comparison table:

TableEdit Scores (EN | ZH):
- PP-StructureV3: 0.159 | 0.109
- Qwen2.5-VL-72B: 0.341 | 0.262
- GOT-OCR: 0.459 | 0.52
- Marker-1.2.3: 0.619 | 0.685
- MinerU-0.9.3: 0.18 | 0.344
- MinerU-1.3.11: 0.1613 | 0.1833
- Mathpix: 0.243 | 0.32
- Gemini2.0-flash: 0.193 | 0.206
- Gemini2.5-Pro: 0.13 | 0.119
- GPT4o: 0.234 | 0.329

Best Performance: PP-StructureV3 achieves the lowest (best) TableEdit scores overall: "0.159" for English and "0.109" for Chinese, making it the single best performer on this metric among all methods listed.

The document confirms "MinerU-0.9.3" appears in the table. No MonkeyOCR, MinerU2.0, or dots.ocr entries are present in this benchmark.

11. user (2026-05-31T04:26:02.108Z)

Web search results for query: "OmniDocBench leaderboard "TableEdit" PP-StructureV3 MinerU2.0 dots.ocr MonkeyOCR best table 2025"

Links: [{"title":"GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub","url":"https://github.com/opendatalab/OmniDocBench"},{"title":"MinerU2.5: A Decoupled Vision-Language Model for Efficient","url":"https://wangbindl.github.io/publications/MinerU2_5.pdf"},{"title":"Changelog - MinerU","url":"https://opendatalab.github.io/MinerU/reference/changelog/"},{"title":"OmniDocBench Leaderboard: PDF Document Parsing SOTA | CodeSOTA | CodeSOTA","url":"https://www.codesota.com/browse/computer-vision/document-parsing/omnidocbench"},{"title":"OmniDocBench Leaderboard: Document Parsing SOTA | CodeSOTA | CodeSOTA","url":"https://www.codesota.com/ocr/benchmark/omnidocbench"},{"title":"dots.ocr: Multilingual Document Layout Parsing in a Single Vision-Language Model","url":"https://arxiv.org/pdf/2512.02498"},{"title":"MinerU2.5: A Decoupled Vision-Language Model for Efficient High-Resolution Document Parsing","url":"https://arxiv.org/html/2509.22186v1"},{"title":"OmniDocBench is Saturated, What?s Next for OCR Benchmarks?","url":"https://www.llamaindex.ai/blog/omnidocbench-is-saturated-what-s-next-for-ocr-benchmarks"},{"title":"MonkeyOCR v1.5 Technical Report: Unlocking Robust Document Parsing for Complex

Patterns","url":"https://arxiv.org/pdf/2511.10390"},{"title":"OmniDocBench 1.5 Benchmark Leaderboard","url":"https://llm-stats.com/benchmarks/omnidocbench-1.5"}]

Based on the search results, here's what I found about the OmniDocBench leaderboard and the models you mentioned:

OmniDocBench Overview

OmniDocBench is a comprehensive document parsing benchmark created by Shanghai AI Laboratory and accepted at CVPR 2025 that evaluates the ability of AI systems to convert PDF documents into structured formats like Markdown, preserving text, tables, formulas, and reading order.

Performance Comparisons

MinerU2.5 vs Other Models

MinerU2.5 consistently outperforms both general-purpose VLMs (e.g., Gemini-2.5 Pro, Qwen2.5-VL-72B, GPT-4o) and domain-specific models (e.g., MonkeyOCR, dots.ocr, PP-StructureV3), establishing new performance records in text recognition, formula recognition, table recognition, and reading order prediction.

dots.ocr Performance

dots.ocr achieved new state-of-the-art scores of 87.5 (EN) and 84.0 (CH).

MonkeyOCR v1.5 Latest Results

MonkeyOCR v1.5 achieves state-of-the-art performance on the widely adopted OmniDocBench v1.5 benchmark, surpassing the previous best-performing methods PPOCR-VL and MinerU 2.5 by 0.15% and 2.34%, respectively, in overall performance.

Current SOTA

Current CodeSOTA SOTA composite is GLM-OCR at 94.62, with composite, table TEDS, formula CDM and text edit-distance scores for leading OCR models.

The search results indicate that MonkeyOCR v1.5 is currently leading on OmniDocBench v1.5, though earlier results showed MinerU2.5 as a top performer.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-05-31T04:26:09.657Z)

The current PaddleOCR page actually DOES still contain "MinerU-0.9.3: 0.18 | 0.344" ? alongside MinerU-1.3.11. So the claim's specific numbers are present in the source. The first fetch's omission of MinerU-0.9.3 was incomplete fetching; the source has both MinerU rows. Now let me assess the two real issues: (1) whether the per-cell facts are accurately quoted, and (2) whether the generalization "outperforms large general-purpose VLMs and pipeline competitors" holds, since search reveals Gemini2.5-Pro (0.13/0.119) beats PP-StructureV3 on this very table, and newer models (MinerU2.5, dots.ocr, MonkeyOCR) claim to beat it.

Let me verify the Gemini number on the page itself, since that's the strongest in-source contradiction to the "outperforms VLMs" framing.

13. user (2026-05-31T04:26:15.247Z)

OmniDocBench Table TEDS Scores

Based on the leaderboard data provided, here's what's available for the requested models:

Available Table TEDS Scores:

- **PaddleOCR-VL**: 93.52 (ranks #1)
- **Qianfan-OCR**: 91.02 (ranks #2)

Models Not Listed:

The following models have no Table TEDS scores in the provided leaderboard:
- PP-StructureV3
- Qwen2.5-VL-72B
- Gemini-2.5-Pro
- MinerU (any version)
- GOT-OCR
- dots.ocr
- MonkeyOCR

Key Finding:

According to the data, "paddleocr-vl" leads with a score of 93.52 on table structure recognition. Neither PP-StructureV3 nor the Vision Language Models you mentioned appear in the Table TEDS rankings, so no direct comparison can be made between them on this specific metric. The leaderboard shows limited table-specific evaluation data overall, with only four models reporting this metric.

14. user (2026-05-31T04:26:17.360Z)

Web search results for query: "PaddleOCR PP-StructureV3 OmniDocBench Gemini-2.5-Pro table edit 0.13 comparison VLM"

Links: [{"title":"PaddleOCR: Efficient OCR Toolkit","url":"https://www.emergentmind.com/topics/paddleocr"},{"title":"GitHub - opendatalab/OmniDocBench: [CVPR 2025] A Comprehensive Benchmark for Document Parsing and Evaluation · GitHub","url":"https://github.com/opendatalab/OmniDocBench"},{"title":"PaddleOCR 3.0 Technical Report","url":"https://arxiv.org/html/2507.05595v1"},{"title":"PaddleOCR-VL 1.5: A Deep Dive into the 0.9B Model That Outperforms GPT-4o on Document Parsing | by Mustafa Genc | Apr, 2026 | Towards AI","url":"https://pub.towardsai.net/paddleocr-vl-1-5-a-deep-dive-into-the-0-9b-model-that-outperforms-gpt-4o-on-document-parsing-c93bac97ac1f?gi=afb181b47933"},{"title":"Releases · PaddlePaddle/PaddleOCR","url":"https://github.com/PaddlePaddle/PaddleOCR/releases"},{"title":"GitHub - PaddlePaddle/PaddleOCR: Turn any PDF or image document into structured data for your AI. A powerful, lightweight OCR toolkit that bridges the gap between images/PDFs and LLMs. Supports 100+ languages. · GitHub","url":"https://github.com/PaddlePaddle/PaddleOCR"},{"title":"A 0.9B Open-Source Model for SOTA Document Parsing: Outperforming GPT-4o and Gemini 2.5 Pro | by Bytefer | Medium","url":"https://medium.com/@bytefer/a-0-9b-open-source-model-for-sota-document-parsing-outperforming-gpt-4o-and-gemini-2-5-pro-f3187827f2c8"},{"title":"PaddleOCR 3.0 Technical Report","url":"https://arxiv.org/pdf/2507.05595"}]

Based on the search results, here's what I found regarding your query about PaddleOCR PP-StructureV3, OmniDocBench, Gemini-2.5-Pro, and table editing performance comparisons:

Key Findings:

PP-OCRv5/PP-StructureV3 reached OmniDocBench edit distances of 0.145 (English) and 0.206 (Chinese), matching Gemini-2.5-Pro's billion-scale VLM accuracy at <100M parameter cost.

These models with fewer than 100 million parameters achieve competitive accuracy and efficiency, rivaling billion-parameter VLMs.

About the Benchmark:

PP-StructureV3 was added to OmniDocBench evaluations as of July 31, 2025, and Gemini-2.5 Pro was added to the benchmark evaluation as of March 27, 2025.

OmniDocBench evaluates four independent dimensions ? text recognition, table reconstruction, formula recognition, and reading order ? and table quality is measured by TEDS (Tree Edit Distance-based Similarity, which captures both structural and content fidelity).

Note: The value 0.13 you mentioned in your query doesn't appear in the search results. The results show edit distance metrics of 0.145 (English) and 0.206 (Chinese) for PP-StructureV3 on OmniDocBench.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-05-31T04:26:17.781Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)